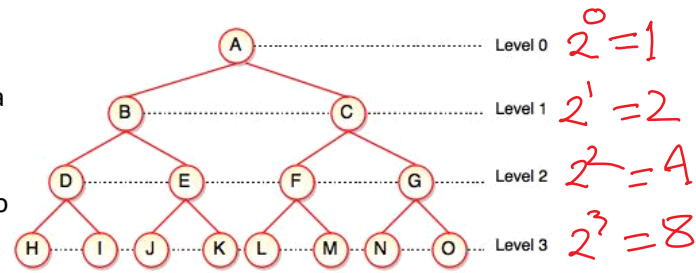


Complete and Extended Binary Tree

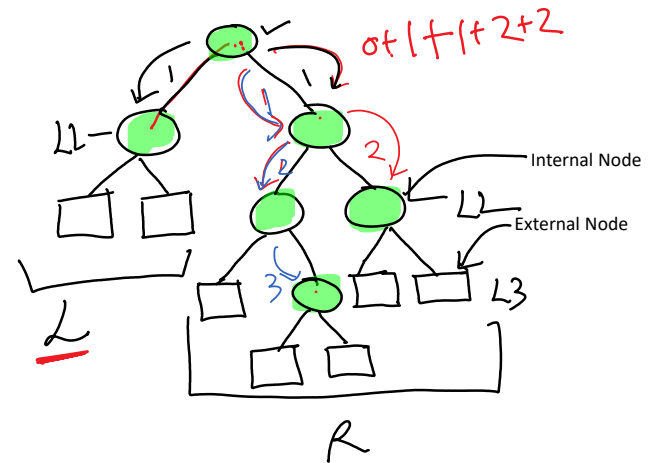
Complete Binary Tree:

1. If all levels of tree are completely filled except the last level and the last level has all keys as left as possible, is said to be a Complete Binary Tree.
2. Complete binary tree is also called as Perfect Binary Tree.
3. In a complete binary tree, every internal node has exactly two children and all leaf nodes are at same level.
4. For example, at Level 2, there must be $2^2 = 4$ nodes and at Level 3 there must be $2^3 = 8$ nodes.



Extended Binary Tree (Also known as 2 Tree):

1. The binary tree where each node has either 0 or two children is called the extended binary tree
2. Nodes with 0 number of children is called the external nodes
3. Nodes with 2 number of children is called the internal nodes
4. If number of internal node is N_I , number of external node is N_E , then $N_E = N_I + 1$
5. Here, $N_I = 6$ and $N_E = 7$
6. Internal path length, $L_I = 0 + 1 + 1 + 2 + 2 + 3 = 9$
7. External path length $L_E = L_I + 2 \times N_I = 21$



$$L_I = 0 + 1 + 1 + 2 + 2 + 3 = 9$$

$$L_E = L_I + 2 \times N_I = 9 + (2 \times 6) = 21$$